Basis for a Topology.

#1 $A = \bigcup_{x \in A} \mathcal{U}_x$.

#3. The Co-Countable is Topology, but $\mathcal{T}_\infty$ is Not Topology.

Counter example.

Let $X = \mathbb{R}$. Let $\mathcal{U}_x = \{x\}$. Then, $\mathcal{U}_x \in \mathcal{T}_\infty$, but for fixed $y \in \mathbb{R}$,

$$\bigcup_{x \neq y} \mathcal{U}_x = \mathbb{R} \setminus \{y\} \notin \mathcal{T}_\infty.$$

That is, $\mathcal{T}_\infty$ is <u>Not</u> closed under the union.

#4). a). Let $\{\mathcal{T}_\alpha\}$ be a family of Topologies on X.

Then, $\bigcap \mathcal{T}_\alpha$ is also Topology on X. But $\bigcup \mathcal{T}_\alpha$ is not.

Proof. Clearly $\emptyset, X \in \bigcap \mathcal{T}_\alpha$. And, let $\mathcal{U}_i \in \bigcap \mathcal{T}_\alpha$ $(i \in I)$ be given.

Then, $\mathcal{U}_i \in \mathcal{T}_\alpha$ for all $i \in I$ and $\alpha \in \Lambda$, thus $\bigcup_{i \in I} \mathcal{U}_i \in \mathcal{T}_\alpha$ for all $\alpha \in \Lambda$,

So it is contained in $\bigcap \mathcal{T}_\alpha$. Similarly, closed under the intersection.

Counter example: $X = \{a, b, c\}$, $\mathcal{T}_1 = \{\emptyset, X, \{a\}, \{a, b\}\}$, $\mathcal{T}_2 = \{\emptyset, X, \{a\}, \{b, c\}\}$,

Then, $\{a, b\} \cap \{b, c\} = \{b\} \notin \mathcal{T}_1 \cup \mathcal{T}_2$

b). Let $S = \{\mathcal{U} \mid \mathcal{U} \in \mathcal{T}_\alpha \text{ for some } \alpha\}$. Then, it is subbasis for X,

and generates the smallest topology on X containing all $\mathcal{T}_\alpha$.

The unique largest topology is $\bigcap \mathcal{T}_\alpha$.

#5. If $\mathcal{A}$ is a basis of $(X, \mathcal{T})$, then $\mathcal{T}_\mathcal{A} = \bigcap_{\substack{\mathcal{A} \subseteq \mathcal{T}' \\ \mathcal{T}' \text{ is topology}}} \mathcal{T}'$

Proof. Let $\mathcal{U} \in \mathcal{T}_\mathcal{A}$ be given.

That, for some $\mathcal{O} \subseteq \mathcal{A}$, $\mathcal{U} = \bigcup \mathcal{O} \in \bigcap \mathcal{T}'$.

The Converse is trivial.

#6. Let $(-1,1)\setminus K \in \mathcal{T}_K$ be given. Then, $(-1,1)\setminus K \notin \mathcal{T}_\ell$, because for $0 \in (-1,1)\setminus K$, there is no basis element of $\mathcal{T}_\ell$ s.t. $0 \in B \subseteq (-1,1)\setminus K$.
Conversely, $[0,1) \in \mathcal{T}_\ell$ is not contained in $\mathcal{T}_K$, because for $0 \in [0,1)$, there is no basis element of $\mathcal{T}_K$ s.t. $0 \in B \subseteq [0,1)$.

#7.

#8. b) The given $\mathcal{C}$ can not generate $[\sqrt{2}, 3)$.
If $[\sqrt{2},3) \in \mathcal{T}_\mathcal{C}$, then for some $a_i \in \mathbb{Q}$ $(i \in \mathbb{Z})$,

$$[\sqrt{2}, 3) = \bigcup_{i \in \mathbb{Z}} [a_i, b_i)$$

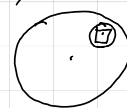
Then, $\sqrt{2} \in [a_j, b_j)$ for some $j \in \mathbb{Z}$, and since $a_j \in \mathbb{Q}$, $a_j < \sqrt{2}$.
Contradiction.

「Subspace.

#5. Let $(X, \mathcal{U}_1)$ and $(X, \mathcal{U}_2)$ be Topological spaces with $\mathcal{U}_1 \subseteq \mathcal{U}_2$
and let $(Y, \mathcal{U}_3)$ and $(Y, \mathcal{U}_4)$ be Topological spaces with $\mathcal{U}_3 \subseteq \mathcal{U}_4$.
Then, $(X, \mathcal{U}_1) \times (Y, \mathcal{U}_3)$ is Coarer than $(X, \mathcal{U}_2) \times (Y, \mathcal{U}_4)$.
a) is trivial. To show the Converse, let $\mathcal{U} \in \mathcal{U}$ be given.
Then, $\mathcal{U} \times Y \subseteq X \times Y$ is open, and $X' \times Y'$ is finer than $X \times Y$, $\mathcal{U} \times Y \subseteq X' \times Y'$ is open.
Now, $\pi[\mathcal{U} \times Y] = \mathcal{U} \subseteq X'$ is open. ☒

#6. Let $B_r(x) \subseteq \mathbb{R}^2$ be given. Then, for any $y \in B_r(x)$, put

$$\varepsilon = r - d(x,y).$$



Finally take $a, b, c, d \in \mathbb{Q}$ s.t.

$y_1 < a < y_1 + \varepsilon/\sqrt{2}$, $y_1 - \varepsilon/\sqrt{2} < b < y_1$
$y_2 < c < y_1 + \varepsilon/\sqrt{2}$, $y_2 - \varepsilon/\sqrt{2} < d < y_2$.

#7. Let $X = \mathbb{R} \setminus \{0\}$, $Y = (0, \infty)$.
Then Y is Convex, but neither ray nor interval.
$(0 \notin X)$

#8. If L is a graph of $x = a$ for some $a \in \mathbb{R}$,
then $L \cong \mathbb{R}$ under the subspace of $\mathbb{R}_\ell \times \mathbb{R}$,
and $L \cong \mathbb{R}_\ell$ 〃 $\mathbb{R}_\ell \times \mathbb{R}_\ell$.
And, in another case, $L \cong \mathbb{R}_\ell$.

#9.

Let $\mathcal{U} \subseteq \mathbb{R}\times\mathbb{R}$ be an basis of $\mathbb{R}^2$ with dic. order.

Let $(x,y) \in \mathcal{U}$ be given. Since $\mathbb{R}_d$ is discrete, $\{x\}$ is open of $\mathbb{R}_d$.

Meanwhile, the $\mathcal{U} = \{((a,b),(c,d)) \mid (a,b) < (c,d)\}$, for this y,

there is $\varepsilon > 0$ s.t $(x, y-\varepsilon) < (x,y) < (x, y+\varepsilon)$ and $(x, y \pm \varepsilon) \in \mathcal{U}$.

Now, $(x,y) \in \{x\} \times (y-\varepsilon, y+\varepsilon) \subseteq \mathcal{U}$, thus $\mathbb{R}_d \times \mathbb{R}$ is finer than $\mathbb{R}^2$, dic.

Conversely, let $\{x\} \times (c,d) \subseteq \mathbb{R}_d \times \mathbb{R}$ be a basis element.

Then, it is clearly open of $\mathbb{R}^2$ dic. order.

#10. $I^2 \neq I^2_d$, because $\{\frac{1}{2}\} \times (\frac{1}{3}, \frac{2}{3})$ is open of $\mathbb{R}_d$, but not open in $\mathbb{R}^2$.

But, any basis of I^2 is open of I^2_d, because

$(y_1, y_2) \in B_r(x) \cap I^2$,

there is $\varepsilon > 0$ s.t $(y_1, y_2) \in ((y_1, y_2 - \varepsilon), (y_1, y_2 + \varepsilon)) \cap I^2 \subseteq B_r(x) \cap I^2$.

Section 17. Closed sets and Limit Points

#2. If A is closed in Y and Y is closed in X, then A is closed in X.

Proof. Y is closed in X $\Leftrightarrow$ $X \setminus Y$ is open of X

A is closed in Y $\Leftrightarrow$ $Y \setminus A$ is open of Y

$\Leftrightarrow$ $Y \setminus A = Y \cap \mathcal{U}$ for some open $\mathcal{U} \subseteq X$

$$X \setminus A = (X \setminus Y) \cup (Y \setminus A) = (X \setminus Y) \cup (Y \cap \mathcal{U}) = (X \setminus Y) \cup \mathcal{U} \text{ is open of } X.$$

#3. If $A \subseteq X$ closed and $B \subseteq Y$ closed, then $A \times B \subseteq X \times Y$ is closed.

Proof. $(X \times Y) \setminus (A \times B) = (X \setminus A) \times Y \cap X \times (Y \setminus B)$

#+ Let Y be a Hausdorff space, X be any space, and $A \subset X$ be a dense in X.

If $f, g : X \to Y$ are continuous and $f|_A = g|_B$, then $f = g$.

Proof. Let $x \in X$ be given. Suppose that $f(x) \neq g(x)$. Then, there are open $\mathcal{U}, V$ set

$$f(x) \in \mathcal{U}, \quad g(x) \in V, \quad \mathcal{U} \cap V = \emptyset.$$

Meanwhile, since f and g both are continuous, $f^{-1}[\mathcal{U}]$ and $g^{-1}[V]$ are opens containing x.

Therefore, $f^{-1}[\mathcal{U}] \cap g^{-1}[V] \ni x$ is non-empty open set, thus

$$f^{-1}[\mathcal{U}] \cap g^{-1}[V] \cap A \neq \emptyset$$

since A is dense in X. Let $a \in f^{-1}[\mathcal{U}] \cap g^{-1}[V] \cap A$. Then,

$$f(a) \in \mathcal{U}, \quad g(a) \in V, \quad a \in A, \text{ thus } f(a) = g(a), \quad \mathcal{U} \cap V \neq \emptyset.$$

It is Contradiction.

#5. $(a,b)\subseteq[a,b]$ is trivial so enough to show that $(a,b)'\subseteq[a,b]$.

Let $\alpha\in X$ be a limit point of (a,b). That is, for any open $\mathcal{U}$ containing α,

$$(\mathcal{U}\setminus\{\alpha\})\cap(a,b)\neq\phi. \quad (*)$$

If $\alpha\notin[a,b]$, that is, $\alpha<a$ or $\alpha>b$. Either $\alpha<a$ or $\alpha>b$, see that:

$\alpha<a \Rightarrow [m,a)$ (if $m\in X$ is the smallest element) contains α, but does not satisfy $(*)$.

$\quad$ " $\Rightarrow (r,a)$ (if such m does not exist) "

And similarly in case of $\alpha>b$.

The equality cannot hold when $X=\mathbb{N}$, $(1,3)$.

#6. a) is clear, by definition.

b) is shown by:

$$x\in\overline{A\cup B} \Leftrightarrow \text{for any open } \mathcal{U} \text{ containing } x,\ \mathcal{U}\cap(A\cup B)\neq\emptyset$$

and since $\mathcal{U}\cap(A\cup B)=(\mathcal{U}\cap A)\cup(\mathcal{U}\cap B)$, either $\mathcal{U}\cap A$ or $\mathcal{U}\cap B$ is non-empty.

So, $x\in\overline{A}$ or $x\in\overline{B}$. The converse is trivial by a).

c). For each β, $A_\beta\subseteq\bigcup A_\alpha$, so $\overline{A_\beta}\subseteq\overline{\bigcup A_\alpha}$.

But the converse is false, by the counterexample : $X=\mathbb{R}$, $A_n=\{\frac{1}{n}\}$. Then,

$$\overline{\bigcup_{n\in\mathbb{N}} A_n}=\{0\}\cup\{\tfrac{1}{n}\mid n\in\mathbb{N}\} \text{ but } \bigcup_{n\in\mathbb{N}}\overline{A_n}=\{\tfrac{1}{n}\mid n\in\mathbb{N}\}.$$

#8. a). $\overline{A\cap B}\subseteq\overline{A}\cap\overline{B}$. Let $X=\mathbb{R}$, $A=(0,1)$, $B=(1,2)$. Then

$$\overline{A\cap B}=\overline{\emptyset}=\emptyset \text{ but } \overline{A}\cap\overline{B}=\{1\}.$$

And, since $\overline{A\cap B}\subseteq\overline{A},\overline{B}$, so $\overline{A\cap B}\subseteq\overline{A}\cap\overline{B}$.

b) has the same result.

c). Let $X=\mathbb{R}$, $A=[0,1)$, $B=(0,1)$. Then

$$\overline{A-B}=\{0\} \text{ but } \overline{A}-\overline{B}=\phi. \text{ So } \overline{A-B}\subset\overline{A}-\overline{B} \text{ does not hold.}$$

And, let $x\in\overline{A}-\overline{B}$ be given. That is, $x\in\overline{A}$ but $x\notin\overline{B}$

Let $\mathcal{U}$ be an open containing x s.t $\mathcal{U}\cap B=\phi$. Since $x\in\overline{A}$, $\mathcal{U}\cap A\neq\phi$.

So, $\mathcal{U}\cap(A-B)=(\mathcal{U}\cap A)\cap(\mathcal{U}\cap B^c)=(\mathcal{U}\cap A)\cap(\mathcal{U}-B)=\mathcal{U}\cap A\neq\phi$

Thus $\overline{A}-\overline{B}\subseteq\overline{A-B}$.

#9.

$(x,y) \in \overline{A} \times \overline{B} \Leftrightarrow$ for any open $U \subseteq A$, $V \subseteq B$ containing x, y, respectively, $U \cap A \neq \phi$ and $V \cap B \neq \phi$.

$\Leftrightarrow$ for any basis element $U \times V \subseteq X \times Y$ containing (x,y), $(U \times V) \cap (A \times B) \neq \phi$

$\Leftrightarrow (x,y) \in \overline{A \times B}$.

#11. Let X and Y be hausdorff spaces.

Given distincts $(x_1, y_1), (x_2, y_2) \in X \times Y$, either $x_1 \neq x_2$ or $y_1 \neq y_2$ must be hold

WLOG, suppose $y_1 \neq y_2$. So, there are opens $U_1, U_2 \subseteq Y$ s.t

$$U_1 \cap U_2 \neq \phi, \quad y_1 \in U_1, \quad y_2 \in U_2.$$

So, $(X \times U_1) \cap (X \times U_2) = X \times (U_1 \cap U_2) = \phi$ and

$(x_i, y_i) \in X \times U_i$ $(i = 1, 2)$. And these are clearly open, by definition.

#12. Let X be a Hausdorff and $A \subseteq X$.

Given two distincts $x, y \in A$, there are opens $U, V \subseteq X$ s.t $x \in U$, $y \in V$, $U \cap V = \phi$.

Now, $(U \cap A) \cap (V \cap A) = \phi$, containing x, y.

#13. X is $T_2 \Leftrightarrow \forall (x_1, x_2) \in X \times X$, $(x_1, x_2) \notin \Delta \Rightarrow \begin{cases} \exists \underbrace{U, V}_{\text{open}} \subseteq X \text{ s.t } (x_1, x_2) \subseteq U \times V \\ U \cap V = \phi \ (\Leftrightarrow (U \times V) \cap \Delta = \phi \\ \qquad \Leftrightarrow U \times V \subseteq \Delta^C \end{cases}$

$\Leftrightarrow \Delta$ is closed.

#14. $x = \lim x_n \Leftrightarrow$ Given open $U \subseteq \mathbb{R}$ containing x, there is $N \in \mathbb{N}$ s.t $n \geq N \Rightarrow x_n \in U$

Since for any open $U \subseteq \mathbb{R}$, $\mathbb{R} \setminus U$ is finite, put $\# \mathbb{R} \setminus U = N$. Then,

x_n must be contained in U, for all $n \geq N$

So, the answer is $\mathbb{R}$.

#19. Def: $Bd\, A = \overline{A} \cap \overline{X-A}$.

a). $Int A \cap Bd A = \phi$.

Proof. Suppose that $Int A \cap Bd A \neq \phi$. Let $x \in Int A \cap Bd A$.

Then, there is an open $\mathcal{U}$ containing x s.t $x \in \mathcal{U} \subseteq A$.

And, since $x \in Bd A = \overline{A} \cap \overline{X-A}$,

$$\mathcal{U} \cap A \neq \phi \quad \text{and} \quad \mathcal{U} \cap (X-A) \neq \phi$$

But $\mathcal{U} \cap (X-A) \neq \phi \iff \mathcal{U} \not\subset A$, so it is Contradiction.

And, since $Int A, Bd A \subseteq \overline{A}$, $Int A \cup Bd A \subseteq \overline{A}$.

Conversely, for $x \in \overline{A}$, suppose that $x \notin Int A$. Then, there is an open $\mathcal{U}$ containing x s.t

$$x \in \mathcal{U} \text{ but } \mathcal{U} \not\subset A$$

$\mathcal{U} \not\subset A \iff \mathcal{U} \cap (X-A) \neq \phi$, so $x \in Bd A$. ▨

b). $Bd A = \phi \iff \overline{A} = int A$.

If A is clopen, $\overline{A} = A = int A$. And, if $\overline{A} = int A$, then

$$A \subseteq \overline{A} = int A, \text{ so } A = int A, \text{ and } \overline{A} = int A \subseteq A, \text{ so } \overline{A} = A$$

「Section 18.

#2. Let $V \subseteq Y$ be an open containing $f(x)$. Then,

$$f^{-1}[V] \text{ is open of } X, \text{ containing } x.$$

Since x is the limit point of A, $f^{-1}[V] \cap A \neq \emptyset$. Now,

$$V \cap f[A] \supseteq f[f^{-1}[V]] \cap f[A] = f[f^{-1}[V] \cap A] \neq \emptyset,$$

So $f(x)$ is the limit point of $f[A]$.

#6. $f : \mathbb{R} \to \mathbb{R} ; x \mapsto \begin{cases} x & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q}. \end{cases}$

It is continuous at only $x=0$.

#10. Let $f : A \to B$ and $g : C \to D$ be continuous maps. Then, a map

$$(f, g) : A \times C \to B \times D ; (a, c) \mapsto (f(a), g(c))$$

is continuous.

Proof. Given a basis element $\mathcal{U} \times V \subseteq B \times D$, $f^{-1}[\pi_A[\mathcal{U} \times V]]$ and $g^{-1}[\pi_B[\mathcal{U} \times V]]$

$$(x, y) \in (f, g)^{-1}[\mathcal{U} \times V] \iff (f(x), g(y)) \in \mathcal{U} \times V \iff f(x) \in \mathcal{U} \text{ and } g(y) \in V \iff (x, y) \in f^{-1}[\mathcal{U}] \times g^{-1}[V]$$

So, given $(x, y) \in (f, g)^{-1}[\mathcal{U} \times V]$, $f(x) \in \mathcal{U}$ and $g(y) \in V$.

Since f and g are continuous, there are open $M \subseteq A$ and open $N \subseteq C$ such that

$$x \in M \text{ and } f[M] \subseteq \mathcal{U}, \quad y \in N \text{ and } g[N] \subseteq V.$$

Therefore, $(x, y) \in M \times N \subseteq f^{-1}[\mathcal{U}] \times g^{-1}[V] = (f, g)^{-1}[\mathcal{U} \times V]$

#11. Let $F: X\times Y \to Z$ be a map.
We say that F is Continuous in each Variable separately if:
for any $y_0 \in Y$, $F_x: X \to Z: x \mapsto F(x, y_0)$ is Continuous, and
for any $x_0 \in X$, $F_y: Y \to Z: y \mapsto F(x_0, y)$ is Continuous.

Show that F is Continuous $\Rightarrow$ F is Continuous in each Variable separately

But the Converse does not hold.
Proof. Let $y_0 \in Y$ be fixed. Then, a map

$$e_x: X \to X\times Y : x \mapsto (x, y_0)$$

is embedding. That is, Continuous, open, one-to-one map.
Continuous: for any basis $U\times V \subseteq X\times Y$, $e_x^{-1}[U\times V] = \begin{cases} U & \text{if } y_0 \in V \\ \emptyset & \text{if } y_0 \notin V \end{cases}$

either $y_0 \in V$ or not, the preimage preserves open
open: for any open $U \subseteq X$, $U \times \{y_0\}$ is the open set of the subspace $X \times \{y_0\}$.

Now, $F_x = e_x \circ F$ is Composition of Continuous maps, so open.
Similarly F_y.

But, there is a Counterexample for the Converse:

$$F: \mathbb{R}^2 \to \mathbb{R} : (x,y) \mapsto \begin{cases} 0 & \text{if } (x,y) = (0,0) \\ \dfrac{xy}{x^2+y^2} & \text{otherwise} \end{cases}$$

Section 19.

$+$. $f: \mathbb{R} \to \mathbb{R}^{\mathbb{N}};\ t \mapsto (t, t, \cdots, t)$ is continuous under the product topology, but not continuous under the box topology.

Proof. Let $A = \prod_{n=1}^{\infty} (-\frac{1}{n}, \frac{1}{n})$ be given. Clearly, A is open of $\mathbb{R}^{\mathbb{N}}$, with box topology.
But, $f^{-1}[A] = \{0\}$, which is not open of $\mathbb{R}$, so f is not continuous.

#6. Let $\{A_n\}_{n \in \mathbb{N}}$ be a sequence in $\prod_{\alpha \in \Lambda} X_\alpha$. Then,

A_n converges to A if and only if for each α, $\{\pi_\alpha(A_n)\}_{n \in \mathbb{N}}$ converges to $\pi_\alpha(A)$.

Proof. A_n converges to A if and only if

for any basis element $B \subseteq \prod X_\alpha$ containing A, there is $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow A_n \in B$

Suppose that $A_n \to A$. Given $\alpha \in \Lambda$, let $U_\alpha \subseteq X_\alpha$ be an open containing $\pi_\alpha(A)$.
Then, $B = \prod_{\alpha \in \Lambda} U_\alpha$ where $U_\beta = X_\beta$ if $\beta \neq \alpha$ is basis element containing A, so

there exists $N \in \mathbb{N}$ s.t. $n \geq N \Rightarrow A_n \in B$. That is, $\pi_\alpha(A_n) \in U_\alpha = \pi_\alpha[B]$, $\forall n \geq N$.
Conversely, let $B \subseteq \prod X_\alpha$ be a basis element containing $A \in B$.
By definition, $B = \prod U_\alpha$ s.t. all $U_\alpha = X_\alpha$ but finitely many $\beta \in \Lambda$.

Let such $\beta \in \Lambda$ as $\beta_1, \cdots, \beta_k$. By the assumption, for each β_i, there is $N_i \in \mathbb{N}$ s.t.

$$n \geq N_i \Rightarrow \pi_{\beta_i}(A_n) \in U_{\beta_i}.$$

Take $N = \max\{N_1, \cdots, N_k\}$. Now,

$$n \geq N \Rightarrow A_n \in B.$$

▨

In box topology, $\Rightarrow$ holds but does not $\Leftarrow$.

#7. $\mathbb{R}^\infty = \bigcup_{n=1}^{\infty} \prod_{k=1}^{n} X_k^n$ where $X_k^n = \begin{cases} \{0\} & k > n \\ \mathbb{R} & k \le n \end{cases}$

$A \in \mathbb{R}^{\mathbb{N}}$ is contained in $\overline{\mathbb{R}^\infty}$ if and only if :

for any basis $\mathcal{U} \subseteq \mathbb{R}^{\mathbb{N}}$ containing A, $\mathcal{U} \cap \mathbb{R}^\infty \neq \phi$.

Under the Product Topology, for given $A \in \mathbb{R}^{\mathbb{N}}$, denote $\mathcal{U} \subseteq \mathbb{R}^{\mathbb{N}}$ as

$$\mathcal{U} = \prod_{n=1}^{\infty} \mathcal{U}_n \quad (\mathcal{U}_n = \mathbb{R} \text{ all } n \in \mathbb{N} \text{ but finitely many } n, \text{ and } \pi_n(A) \in \mathcal{U}_n, \forall n \in \mathbb{N}.)$$

Then, $$\mathcal{U} \cap \mathbb{R}^\infty = \left(\prod_{n=1}^{\infty} \mathcal{U}_n\right) \cap \left(\bigcup_{n=1}^{\infty}\left(\left[\prod_{k=1}^{n} \mathbb{R}\right] \times \left[\prod_{k=n+1}^{\infty} \{0\}\right]\right)\right)$$

$$= \bigcup_{n=1}^{\infty}\left[\left(\prod_{k=1}^{\infty} \mathcal{U}_k\right) \cap \left(\prod_{k=1}^{\infty} X_k^n\right)\right] \text{ where } X_k^n = \begin{cases} \{0\} & k > n \\ \mathbb{R} & k \le n \end{cases}$$

$$= \bigcup_{n=1}^{\infty} \prod_{k=1}^{\infty} (\mathcal{U}_k \cap X_k^n)$$

It cannot be empty set; because, if $m \in \mathbb{N}$ is a number s.t. $k \ge m \Rightarrow \mathcal{U}_k = \mathbb{R}$, then

$$\prod_{k=1}^{\infty} \mathcal{U}_k \cap \prod_{k=1}^{\infty} X_k^m \neq \phi$$

Thus $\overline{\mathbb{R}^\infty} = \mathbb{R}^{\mathbb{N}}$. But, under the box topology, it cannot be held, for

$$\mathcal{U} = \prod_{n=1}^{\infty} (0,1)$$

valid as counterexample. Actually, $\overline{\mathbb{R}^\infty} = \mathbb{R}^\infty$. Because no such $A \in \overline{\mathbb{R}^\infty}$ with $A \notin \mathbb{R}^\infty$.

#8. h is bijective, with the inverse as

$$h^{-1} : \mathbb{R}^{\mathbb{N}} \to \mathbb{R}^{\mathbb{N}} : (x_n)_{n \in \mathbb{N}} \mapsto \left(\frac{1}{a_n} \cdot [x - b_n]\right)$$

And, regardless of box or Prod Homeo.

#10. Subbasis with Initial Topology.

Let X be a set, and $\{X_\alpha\}_{\alpha\in\Lambda}$ be a Collection of spaces.

1. Given $f_\alpha : X \to X_\alpha$ $(\alpha\in\Lambda)$, there exists a unique Coarsest Topology or $\mathcal{T}$ on X s.t

for all $\alpha\in\Lambda$, f_α is Continuous map.

Proof. Let $S_\alpha \overset{\text{def}}{=} \{f_\alpha^{-1}[U_\alpha] \mid U_\alpha \subseteq X_\alpha \text{ is open of } X_\alpha\}$

Then, $\mathcal{S} \overset{\text{def}}{=} \bigcup_{\alpha\in\Lambda} S_\alpha$ is a subbasis of X, because $X = f_\alpha^{-1}[X_\alpha]$, for any $\alpha\in\Lambda$.

Given $\alpha\in\Lambda$, and open $U_\alpha \subseteq X_\alpha$, $f_\alpha^{-1}[U_\alpha] \in S_\alpha \subseteq \mathcal{S}$, So all f_α are Continuous.

Moreover, it is the Coarest Topology s.t all f_α are Continuous, because

if $\mathcal{T}$ is a Topology on X satisfying such Condition,

then for any $\alpha\in\Lambda$ and open $U_\alpha \subseteq X_\alpha$, $f_\alpha^{-1}[U_\alpha]$ is Contained in $\mathcal{T}$,

therefore $\mathcal{T}_\mathcal{S} \subseteq \mathcal{T}$.

Note: Such a Topology on X is called the initial Topology for $\{f_\alpha\}$.

2. A map $g : Y \to X$ is Continuous iff $f_\alpha \circ g$ is Continuous for all $\alpha\in\Lambda$.

(The given Y is any Space, and X is the initial Topology.)

Proof. Since the Composition of Continuous maps is Continuous, $\Rightarrow$ is trivial.

Conversely, By the assumption, for each $\alpha\in\Lambda$, and an open $U_\alpha \subseteq X_\alpha$,

$$(f_\alpha\circ g)^{-1}[U_\alpha] = g^{-1}[f_\alpha^{-1}[U_\alpha]] \text{ is an open of } Y.$$

Meanwhile, by the first assertion, every open of X is generated by $\mathcal{S}$,

$$g^{-1}[U] \text{ is open for any open } U \subseteq X.$$

「Section. Metric.

#2. Define $d : \mathbb{R}^2 \times \mathbb{R}^2 \to \mathbb{R} : ((x_1, y_1), (x_2, y_2)) \mapsto \begin{cases} |y_1 - y_2| & \text{if } x_1 = x_2 \\ 1 + |y_1 - y_2| & \text{if } x_1 \neq x_2. \end{cases}$

Then, for any open set $\mathcal{U}$ of $\mathbb{R} \times \mathbb{R}$ with the dictionary order Topology, let $(x, y) \in \mathcal{U}$ be given. Then, there are $(x_1, y_1), (x_2, y_2) \in \mathcal{U}$ s.t.

$$(x_1, y_1) < (x, y) < (x_2, y_2)$$

Take $r = \min\{|y_1 - y|, |y_2 - y|\}$. Then, $B_r(x, y) \subseteq \mathcal{U}$.

Conversely, let $B_r(x, y)$ be given as arbitrary open ball.

Take $y_1 = y - \frac{r}{2}$ and $y_2 = y + \frac{r}{2}$. Then,

$$((x, y_1), (x, y_2)) \subseteq B_r(x, y)$$

$\boxtimes$

#.3. Let (X,d) be a Metric Space. Then, the Metric Topology generated by

$$\beta_d := \{B_r(x) \mid r>0,\ x\in X\}$$

is the Coarsest Topology such that $d: X\times X \to \mathbb{R}$ is Continuous.

Proof. Given a basis element $(a,b)\subset\mathbb{R}$ $(a<b)$,

$$d^{-1}[(a,b)] = \{(x,y)\in X\times X \mid d(x,y)\in(a,b)\}.$$

If $b\leq 0$, then the preimage above is empty, so open. So, assume that $b>0$.
Given $(x,y)\in d^{-1}[(a,b)]$, $a<d(x,y)<b$. Take $r_x, r_y > 0$ such that

$$r_x + r_y < \min\{d(x,y)-a,\ b-d(x,y)\}.$$

Now,

$$(x,y)\in B_{r_x}(x)\times B_{r_y}(y) \subseteq d^{-1}[(a,b)],$$ because for any $(x',y')\in B_{r_x}(x)\times B_{r_y}(y)$,

$d(x,x')<r_x$ and $d(y,y')<r_y$, so

$$d(x',y') \leq d(x',x) + d(x,y) + d(y,y') < r_x + d(x,y) + r_y < b$$

and

$$d(x,y) \leq d(x,x') + d(x',y') + d(y',y) < r_x + d(x',y') + r_y,$$ so

$$a < d(x,y) - (r_x+r_y) < d(x',y').$$

That is, $d(x',y')\in(a,b)$, or $(x',y')\in d^{-1}[(a,b)]$.

To show that the second assertion, let $\mathcal{T}$ be a Topology on X such that

$$d: X\times X \to \mathbb{R}$$

is Continuous. Since every Continuous map on the Product Topology is Continuous each variable, given $x_0\in X$, $f_{x_0}: X\to\mathbb{R}; x\mapsto d(x,x_0)$ is Continuous on $(X,\mathcal{T})$.
We want to show that $\mathcal{T}_d\subseteq\mathcal{T}$, so, let $B_r(x_0)\in\mathcal{T}_d$ be given.
for any $y\in B_r(x_0)$, $B_\epsilon(y)\subseteq B_r(x_0)$, where $\epsilon = r - d(x_0,y)$.
Meanwhile, $B_\epsilon(y) = f_y^{-1}[(-\infty,\epsilon)]$, so $B_r(x_0)$ is open of $(X,\mathcal{T})$.
Thus Proved.

#4.

$f: \mathbb{R} \to \mathbb{R}^{\mathbb{N}};\ t \mapsto (t, 2t, 3t, \cdots)$

$g: \mathbb{R} \to \mathbb{R}^{\mathbb{N}};\ t \mapsto (t, t, \cdots)$

$h: \mathbb{R} \to \mathbb{R}^{\mathbb{N}};\ t \mapsto (t, \frac{1}{2}t, \frac{1}{3}t, \cdots)$

Continuous table

map \ T.P	Product	box	Uniform
f	O	X	X
g	O	X	O
h	O	X	O

(1,1): the preimage is finite intersection of opens.

(2,1): 〃

(3,1): 〃

But, in the box topology, let $\mathcal{U} = \prod_{n=1}^{\infty} \left(-\frac{1}{n^2}, \frac{1}{n^2}\right)$. Then,

$$nt \in \left(-\frac{1}{n^2}, \frac{1}{n^2}\right) \Leftrightarrow t \in \left(-\frac{1}{n^3}, \frac{1}{n^3}\right) \text{ and}$$

$$t \in \left(-\frac{1}{n^2}, \frac{1}{n^2}\right)$$

$$\frac{1}{n}t \in \left(-\frac{1}{n^2}, \frac{1}{n^2}\right) \Leftrightarrow t \in \left(-\frac{1}{n}, \frac{1}{n}\right)$$

So, each f, g, h, the preimages are $\{0\}$, closed. Thus Not continuous.

Recall: (X, d), $\bar{d}: X \times X \to \mathbb{R}: (x, y) \mapsto \min\{d(x,y), 1\}$

$(\mathbb{R}^{\mathbb{N}}, d)$. $\rho: \mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}} \to \mathbb{R}: (\{x_n\}, \{y_n\}) \mapsto \sup\{\bar{d}(x_n, y_n) \mid n \in \mathbb{N}\}$.

Given $x \in \mathbb{R}$ and $\epsilon > 0$, take $\delta = \frac{\epsilon}{2}$

Then, $|x-y| < \delta \Rightarrow \rho(f(x), f(y)) = \sup_{n \in \mathbb{N}} \min\{|nx - ny|, 1\}$ (false)

$|x-y| < \delta \Rightarrow \rho(g(x), g(y)) = \sup_{n \in \mathbb{N}} \min\{|x-y|, 1\} < \epsilon$

$|x-y| < \delta \Rightarrow \rho(h(x), h(y)) = \sup_{n \in \mathbb{N}} \min\{\frac{1}{n}|x-y|, 1\} < \epsilon$ ∎

Sec 20.5.

Let $\mathbb{R}^\infty = \bigcup_{n=1}^{\infty} \underbrace{\mathbb{R}\times\mathbb{R}\times\cdots\times\mathbb{R}}_{n}\times\{0\}\times\{0\}\times\cdots \subseteq \mathbb{R}^{\mathbb{N}}$.

What is $\overline{\mathbb{R}^\infty}$ under the Uniform Topology?

Proof. $\mathbb{R}^\infty \subseteq \overline{\mathbb{R}^\infty}$ is clear, just the property of closure.

Let $\{x_n\} \in \mathbb{R}^{\mathbb{N}}$ be a sequence s.t. $x_n \to 0$ as $n\to\infty$.

Then, given $\varepsilon>0$, there is $N\in\mathbb{N}$ s.t. $n\ge N \Rightarrow |x_n|<\varepsilon$ or $0\in B_\varepsilon(x_n)$.

That is, given $\varepsilon>0$, $\mathbb{R}^\infty \cap B_{2\varepsilon}(\{x_n\})$ is not empty,

for fixed $\underbrace{\mathbb{R}\times\mathbb{R}\times\cdots\times\mathbb{R}}_{N}\times\{0\}\times\cdots$, $x_1,\cdots,x_{N-1}\in\mathbb{R}$, and $|x_k|<\varepsilon$ for all $k\ge N$, so

$$(x_1, x_2, \cdots, x_{N-1}, 0, 0, \cdots) \in \mathbb{R}^\infty \cap B_{2\varepsilon}(\{x_n\}).$$

So, $\overline{\mathbb{R}^\infty} = \{(x_n) \mid x_n \to 0 \text{ as } n\to\infty\}$,

(If $\{x_n\} \not\to 0$, then we can take $\varepsilon>0$ s.t. $\mathbb{R}^\infty \cap B_\varepsilon(\{x_n\}) = \phi$.).

More rigorously, there is $\varepsilon>0$ s.t. for any $N\in\mathbb{N}$, there is $n\ge N$ s.t. $|x_n|\ge\varepsilon$.

So, for any $\mathbb{R}^{\mathbb{N}}\times\{0\}^\infty$, $(\mathbb{R}^{\mathbb{N}}\times\{0\}^\infty)\cap B_\varepsilon(\{x_n\}) = \phi$, so $\mathbb{R}^\infty \cap B_\varepsilon(\{x_n\}) = \bigcup \phi = \phi$.

20.6.

Let $\bar{\rho}$ be the uniform metric on $\mathbb{R}^{\mathbb{N}}$.
Given $(x_n) \in \mathbb{R}^{\mathbb{N}}$, and $0 < \varepsilon < 1$, define

$$\mathcal{U}((x_n), \varepsilon) \overset{\text{def}}{=} \prod_{n=1}^{\infty} (x_n - \varepsilon, x_n + \varepsilon)$$

a). $\mathcal{U}((x_n), \varepsilon) \neq B_{\bar{\rho}}((x_n), \varepsilon)$. $\bar{\rho} : \mathbb{R}^{\mathbb{N}} \times \mathbb{R}^{\mathbb{N}} \to \mathbb{R} : ((x_n), (y_n)) \mapsto \sup_{n \in \mathbb{N}} \min\{|x_n - y_n|, 1\}$.

b). $\mathcal{U}((x_n), \varepsilon)$ is Not open.

c). $B_{\bar{\rho}}((x_n), \varepsilon) = \bigcup_{\delta < \varepsilon} \mathcal{U}((x_n), \delta)$.

Proof. Let $y_n = x_n + \varepsilon \cdot \frac{n-1}{n}$. Then, for all $n \in \mathbb{N}$, $|x_n - y_n| = \varepsilon \cdot \frac{n-1}{n} < \varepsilon$, so $(y_n) \in \mathcal{U}((x_n), \varepsilon)$. but! $\bar{\rho}((x_n), (y_n)) = \sup_{n \in \mathbb{N}} \varepsilon \cdot \frac{n-1}{n} = \varepsilon$, so $(y_n) \notin B_{\bar{\rho}}((x_n), \varepsilon)$.

As see above, $(y_n) \in \mathcal{U}((x_n), \varepsilon)$. Suppose that $\mathcal{U}((x_n), \varepsilon)$ is open. Then, there is $\delta > 0$ s.t.

$$(y_n) \in B_{\bar{\rho}}((y_n), \delta) \subseteq \mathcal{U}((x_n), \varepsilon).$$

Take $y_n' = y_n + \frac{\delta}{2}$. Then, $(y_n') \in B_{\bar{\rho}}((y_n), \delta)$ but

$$y_n' = y_n + \frac{\delta}{2} = x_n + \frac{n-1}{n} \cdot \varepsilon + \frac{\delta}{2} \notin (x_n - \varepsilon, x_n + \varepsilon)$$

for some enough large $n \in \mathbb{N}$, so contradiction.
(specifically, take $n \in \mathbb{N}$ s.t. $\frac{n-1}{n} \cdot \varepsilon + \frac{\delta}{2} \geq \varepsilon$, or $\frac{\delta}{2} \geq \frac{\varepsilon}{n}$.).

Given $\delta < \varepsilon$, $\mathcal{U}((x_n), \delta) \subseteq B_{\bar{\rho}}((x_n), \varepsilon)$ is trivial, so $B_{\bar{\rho}}((x_n), \varepsilon) \supseteq \bigcup_{\delta < \varepsilon} \mathcal{U}((x_n), \delta)$.
Now, let $(y_n) \in B_{\bar{\rho}}((x_n), \varepsilon)$ be given. Then, $\sup_{n \in \mathbb{N}} |x_n - y_n| < \varepsilon$, so for some $\delta < \varepsilon$, for all $n \in \mathbb{N}$, $|x_n - y_n| \leq \sup_{n \in \mathbb{N}} |x_n - y_n| < \delta < \varepsilon$. So proved. ∎

20, 8. ℓ^2 - Space.

Define a Metric on $X \overset{\text{def}}{=} \{(x_n) \in \mathbb{R}^{\mathbb{N}} \mid \sum_{n=1}^{\infty} x_n^2 < \infty\}$ as

$$d : X \times X \to \mathbb{R} ; ((x_n), (y_n)) \mapsto \sqrt{\sum_{i=1}^{\infty} (x_i - y_i)^2} .$$

The Topological Space (X, d) is called ℓ^2 - Space.

a). Show that on X,

$$\text{box topology} \supset \ell^2\text{- topology} \supset \text{uniform topology},$$

Proof.

[Sec 21.

#21.1. Let $A \subset X$. If $d: X \times X \to \mathbb{R}$ is a Metric for the topology of X, Show that $d|_{A\times A} : A\times A \to \mathbb{R} : (a,b) \mapsto d(a,b)$ is a Metric for the subspace Topology on A.

Proof. Let $(a,b) \subset \mathbb{R}$ be given. Then,

$$d|_{A\times A}^{-1}[(a,b)] = d^{-1}[(a,b)] \cap A\times A$$

so, by definition of subspace, it is continuous.

#21.2. Let (X, d_X) and (Y, d_Y) be Metric spaces.
Let $f: X \to Y$ be a map satisfying

$$d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2) \quad \text{for all } x_1, x_2 \in X.$$

Then, f is an imbedding.

Proof. 1). f is continuous

Let $x \in X$ be fixed. Given $\varepsilon > 0$, let $\delta = \varepsilon$. Then,

$$d_X(x,t) < \delta \Rightarrow d_Y(f(x), f(t)) = d_X(x,t) < \delta = \varepsilon, \text{ so it is continuous.}$$

2). f is one-to-one.

$f(x_1) = f(x_2) \Leftrightarrow d_Y(f(x_1), f(x_2)) = 0 \Leftrightarrow d_X(x_1, x_2) = 0 \Leftrightarrow x_1 = x_2$.

3). f is open map.

Let $B_r(x) \subseteq X$ be given. Then, $f[B_r(x)]$ is open in $f[X]$, because

$f(t) \in f[B_r(x)] \Rightarrow$ for $\varepsilon = r - d(x,t)$, $f(t) \subseteq B_\varepsilon(f(t)) \subseteq f[B_r(x)]$.

Topological Group.

Def. Let G be a Group under the binary operation $\cdot : G\times G \to G$.
A Topological Space $(G, \mathcal{U})$ satisfying T_1 condition, and

$$G\times G \to G : (x,y) \mapsto x\cdot y \quad \text{and} \quad G\to G : x \mapsto x^{-1} \text{ are Continuous}$$

is called the Topological Group.

Lemma. Let G be a group and T_1-space. Then
G is Topological Group if and only if $G\times G \to G : (x,y) \mapsto xy^{-1}$ is Continuous.

Proof. Suppose that G is Topological Group. Then
$\varphi(x,y) = xy$ and $\psi(y) = y^{-1}$ are continuous, therefore

$(x,y) \mapsto xy^{-1} = \varphi(x, \psi(y))$ is continuous, being composition of continuous maps

Conversely, suppose that $\mu : G\times G \to G ; (x,y) \mapsto xy^{-1}$ is Continuous.
Then, for fixed $x=1$, $\mu(1,y) = y^{-1}$ is continuous, so $x \mapsto x^{-1}$ is continuous.
Now, $y \mapsto y^{-1}$ is continuous, so

$$(x,y) \mapsto x\cdot(y^{-1})^{-1} = xy \text{ is continuous, being composition of continuous.}$$

∎

⌈Examples.

⌈$(\mathbb{Z}, +)$ is clearly a Topological Group, because the Topology is discrete.

⌈$(\mathbb{R}, +)$ is Topological Group under the Standard Topology.

Proof. T_1 is clear, by Metric $\Rightarrow T_1$

Let $\varepsilon > 0$ be given. Then, for $\delta = \varepsilon$,

$$\sqrt{(x_1-x_2)^2+(y_1-y_2)^2} < \delta \Rightarrow |x_1-y_1-(x_2-y_2)| = |(x_1-x_1)-(y_1-y_2)| \leq |x_1-x_2|+|y_1-y_2| < 2\varepsilon.$$

Thus, it is a Topological Group.

⌈$(\mathbb{R}_+, \cdot)$ is Topological Group.

Proof. Given $\varepsilon > 0$,

$$\sqrt{(x_1-x_2)^2+(y_1-y_2)^2} < \delta \Rightarrow \left|\frac{x_1}{y_1}-\frac{x_2}{y_2}\right| = \frac{1}{|y_1 y_2|}\cdot|x_1 y_2 - x_2 y_1|$$

「$GL_n(\mathbb{R})$.

Consider $GL_n(\mathbb{R})$ as the subset of $\mathbb{R}^{n^2}$, with Euclidean space.

Define a Metric for $M_{n\times n}(\mathbb{R})$ as;

$$d; M_{n\times n}(\mathbb{R}) \times M_{n\times n}(\mathbb{R}) \to \mathbb{R} : (A, B) \mapsto \sqrt{\sum_{i,j} (A_{ij} - B_{ij})^2}$$

Since the Matrix multiplication

$$\mu; M_{n\times n}(\mathbb{R}) \times M_{n\times n}(\mathbb{R}) \to M_{n\times n}(\mathbb{R}) : (A, B) \mapsto \left(\sum_{k=1}^{n} A_{ik} \cdot B_{kj}\right)_{ij}$$

is Continuous, because it is entry-wise Continuous.

Since $GL_n(\mathbb{R})$ is closed under the multiplication, it has also Continuous operation.

And,

$$GL_n(\mathbb{R}) \to GL_n(\mathbb{R}) : A \mapsto A^{-1}$$

is also Continuous, because $A^{-1} = (\det A)^{-1} \cdot \operatorname{adj} A$

where $(\operatorname{adj} A)_{ij} = (-1)^{i+j} \cdot \det A(j|i)$.

is also multivariables polynomial, entrywise.

「Consider $O_n(\mathbb{R}) = \{A \in M_{n\times n}(\mathbb{R}) \mid A^t = A^{-1}\}$, $SO_n(\mathbb{R}) = \{A \in O_n(\mathbb{R}) \mid \det A = 1\}$.

Then, O_n and SO_n are Compact, and SO_n is connected.

proof. $A \in O_n(\mathbb{R}) \Leftrightarrow A^t A = I \Leftrightarrow \sum_{k=1}^{n} A_{ik} \cdot A_{jk} = \delta_{ij}$.

Define a map $f_{ij}; M_{n\times n}(\mathbb{R}) \to \mathbb{R} : A \mapsto \sum_{k=1}^{n} A_{ik} \cdot A_{jk}$.

Then, it is Continuous therefore

$$O_n(\mathbb{R}) = \bigcap_{\substack{1\le i,j\le n \\ i\ne j}} f^{-1}[\{0\}] \cap \bigcap_{1\le i=j\le n} f^{-1}[\{1\}]$$ is finite intersection of closed set,

thus closed. Moreover, $\operatorname{diam} O_n(\mathbb{R}) \le n$, so it is bounded. Therefore, Compact.

And,

$\det ; O_n(\mathbb{R}) \to \mathbb{R} ; A \mapsto \det A$ is Continuous, So,

$SO_n(\mathbb{R}) = \det^{-1}[O_n(\mathbb{R})]$ also closed, and bounded by subset of bounded set.

「Compact space.

1, a). Let $\mathcal{T}, \mathcal{T}'$ be Topologies on X, with $\mathcal{T} \subseteq \mathcal{T}'$.

If $(X, \mathcal{T}')$ is Compact, then $(X, \mathcal{T})$ is Compact.

Prof. Let $\mathcal{O} \subseteq \mathcal{T}$ be a open Cover of X. Then, since $\mathcal{O} \subset \mathcal{T} \subseteq \mathcal{T}'$ and $(X, \mathcal{T}')$ Compact, $\mathcal{O}$ has finite subcover. ☒

1, b). If X is Compact T_2 under both $\mathcal{T}$ and $\mathcal{T}'$,

then either $\mathcal{T}$ and $\mathcal{T}'$ are equal or they are Not comparable.

Prof. If $\mathcal{T} = \mathcal{T}'$, then there is nothing to prove. Suppose $\mathcal{T} \neq \mathcal{T}'$.

Enough to show that neither $\mathcal{T} \subset \mathcal{T}'$ nor $\mathcal{T}' \subset \mathcal{T}$.

Suppose that $\mathcal{T} \subset \mathcal{T}'$. Then, we can choose an open $\mathcal{U} \in \mathcal{T}' \setminus \mathcal{T}$.

Now, $X \setminus \mathcal{U}$ is closed in $\mathcal{T}'$ but not $\mathcal{T}$. Since $\mathcal{T}'$ is Compact T_2, $X \setminus \mathcal{U}$ is Compact set of $\mathcal{T}'$. Since $\mathcal{T} \subset \mathcal{T}'$, for any open cover $\mathcal{O} \subseteq \mathcal{T}$ for $X \setminus \mathcal{U}$, it has finite subcover. Therefore $X \setminus \mathcal{U}$ is compact in $(X, \mathcal{T})$, thus Closed.

But, it is Contradiction. Similarly, $\mathcal{T}' \not\subset \mathcal{T}$.

2, a). Every subspace in Co-finite Topology $(\mathbb{R}, \mathcal{T}_f)$ is Compact.

b). In Co-Countable $(\mathbb{R}, \mathcal{T}_c)$, is $[0,1]$ Compact?

Prof. Let $A \subseteq \mathbb{R}$ be a subset. Let $\mathcal{O} = \{\mathbb{R} \setminus F \mid F \text{ is finite}\}$ be an open Cover for A. That is,

$$A \subseteq \bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U} . \Leftrightarrow \bigcap_{\mathcal{U} \in \mathcal{O}} \mathbb{R} \setminus \mathcal{U} \subseteq A^c$$

But since $\mathbb{R} \setminus \mathcal{U}$ is finite for each $\mathcal{U} \in \mathcal{O}$, there are finite number of $\mathcal{U}_1, \cdots, \mathcal{U}_n$ s.t

$$\bigcap_{i=1}^{n} \mathbb{R} \setminus \mathcal{U}_i \subseteq A^c \Leftrightarrow A \subseteq \bigcup_{i=1}^{n} \mathcal{U}_i$$

Therefore A is Compact. (More specifically, for any distinct finite set F_1 and F_2, $F_1 \cap F_2$ has number of less than $\min\{\#(F_1), \#(F_2)\}$.)

But, in the Co-Countable, $[0,1]$ is Not compact, because

Construct sequences; denote $\mathbb{Q} \cap [0,1] = \{x_n \mid n \in \mathbb{N}\}$, and let $\mathcal{U}_n = (\mathbb{R} \setminus \mathbb{Q}) \cup \{x_n\}$

Then, $\bigcup_{n=1}^{\infty} \mathcal{U}_n \supseteq [0,1]$ but it has no finite subcover.

⌜3. Finite union of Compact Subspace is Compact.

Proof. Let $K_1, \cdots, K_n \subseteq X$ be Compact Subsets. Let $\mathcal{O} \subseteq \mathcal{P}(X)$ be an open cover of $\bigcup_{i=1}^{n} K_i$

Since $\bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U} \supseteq \bigcup_{i=1}^{n} K_i \supseteq K_j \ (1 \leq j \leq n)$, for each j, $\mathcal{O}$ has a finite subcover for K_j,

denote $\mathcal{U}_j^1, \cdots, \mathcal{U}_j^{r_j} \in \mathcal{O}$. Now, clearly

$$\bigcup_{i=1}^{n} K_i \subseteq \bigcup_{j=1}^{n} \bigcup_{m=1}^{r_j} \mathcal{U}_j^m$$

⌜4. Every Compact Subspace in a Metric space is closed bounded.

Give the example that the Converse fails.

Proof. Closed is given by : Compact in T_2 is closed, because

Let X be a Hausdorff Space, and $K \subseteq X$ be a Compact Subset.

Let $x \in X \setminus K$ be fixed. Now, for each $y \in K$, there exists disjoint open $\mathcal{U}_y^x, \mathcal{U}_y$ s.t

$$x \in \mathcal{U}_y^x, \quad y \in \mathcal{U}_y, \quad \mathcal{U}_y^x \cap \mathcal{U}_y$$

Now, $\{\mathcal{U}_y \mid y \in K\}$ is open Cover of K, therefore there is a finite subcover

$\{\mathcal{U}_{y_i} \mid i = 1, 2, \cdots, r_x\}$ and open neiborhoods of x $\{\mathcal{U}_{y_i}^x \mid i = 1, \cdots, r_x\}$.

Put $V_x = \bigcap_{i=1}^{r_x} \mathcal{U}_{y_i}^x$. Then, V_x is open neiborhood and $K \cap V_x = \phi$, thus $V_x \subseteq K^c$.

(More Specifically, $K \subseteq \bigcup_{i=1}^{r_x} \mathcal{U}_{y_i}$ and $\left(\bigcup_{i=1}^{r_x} \mathcal{U}_{y_i}\right) \cap V_x = \bigcup_{i=1}^{r_x} (\mathcal{U}_{y_i} \cap V_x) = \bigcup_{i=1}^{r_x} \phi = \phi$.

And boundedness is clear.

Counterexample: Let $X = \{\frac{1}{n} \mid n \in \mathbb{N}\}$ discrete Topology.

Then, it is closed itself, and bounded, but not Compact.

「5. Let X be a Hausdorff, and $A, B \subseteq X$ be disjoint compact subsets.
Then, there exist disjoint open sets $\mathcal{U}$ and V containing A and B, respectively.

Proof. Let $x \in A$ be fixed. Then, for each $y \in B$, there exist opens $\mathcal{U}_{(x,y)}$ and $V_{(x,y)}$ s.t

$$x \in \mathcal{U}_{(x,y)},\ y \in V_{(x,y)},\ \mathcal{U}_{(x,y)} \cap V_{(x,y)} = \phi.$$

Since $\{V_{(x,y)} \mid y \in B\}$ is open cover of B, there exist a finite subcover $\{V_{(x,y_i)} \mid i = 1, 2, \cdots, r_x\}$. Now, $\mathcal{U}_x = \bigcap_{i=1}^{r_x} \mathcal{U}_{(x,y_i)}$ is open neiborhood of x, $V_x = \bigcup_{i=1}^{r_x} V_{(x,y_i)}$ is open set containg B, and $\mathcal{U}_x \cap V_x = \phi$. because

$$\mathcal{U}_x \cap V_x = \mathcal{U}_x \cap \left(\bigcup_{i=1}^{r_x} V_{(x,y_i)}\right) = \bigcup_{i=1}^{r_x}\left(\mathcal{U}_x \cap V_{(x,y_i)}\right) = \bigcup_{i=1}^{r_x} \phi = \phi$$

In summary, for each $x \in A$, there are two disjoint opens $\mathcal{U}_x, V_x$ s.t

$$x \in \mathcal{U}_x,\ B \subseteq V_x,\ \mathcal{U}_x \cap V_x = \phi.$$

And, $\{\mathcal{U}_x \mid x \in A\}$ is an open subcover of A, thus it has a finite subcover $\{\mathcal{U}_{x_i} \mid i = 1, \cdots, m\}$. Finally,
$\bigcup_{i=1}^{m} \mathcal{U}_{x_i}$ and $\bigcap_{i=1}^{m} V_{x_i}$ are disjoint open sets containing A and B, respectively.
(Disjointness is given by $\left(\bigcup_{i=1}^{m} \mathcal{U}_{x_i}\right) \cap \left(\bigcap_{i=1}^{m} V_{x_i}\right) = \bigcup_{i=1}^{m}\left(\mathcal{U}_{x_i} \cap \left(\bigcap_{i=1}^{m} V_{x_i}\right)\right) = \bigcup \phi = \phi.$

「6. If $f: X \to Y$ is continuous map where X is compact and Y is Hausdorff, then f is closed map.

Proof. Let $C \subseteq X$ be a closed set. Since X is compact, C is compact.
And, the continuous map f preserves compactness, $f[C]$ is compact in T_2, thus closed.

Γ7. If Y is Compact, then the projection $P_X : X\times Y \to X$ is a Closed map.

Proof. Let $C \subseteq X\times Y$ be a Closed set. Want to Show that $X \setminus P_X[C]$ is open.
Let $x \in X \setminus P_X[C]$ be given. Then, $(\{x\}\times Y) \cap C = \phi$, therefore $\{x\}\times Y \subseteq (X\times Y)\setminus C$.
Now, by the Tube Lemma, there is an open $\mathcal{U} \subseteq X$ s.t

$$\{x\}\times Y \subseteq \mathcal{U}\times Y \subseteq (X\times Y)\setminus C.$$

Now, $x \in \mathcal{U}$ and $\mathcal{U} \subseteq X\setminus P_X[C]$, thus $X \setminus P_X[C]$ is open.
Note: the Converse is true, but so hard to prove.

ΓRecall: The Tube Lemma.
Let X be a Topological Space, and Y be a Compact Space.
Then, for the product space $X\times Y$, for fixed $x_0 \in X$,

For any open $N \subseteq X\times Y$ s.t $\{x_0\}\times Y \subseteq N$, there is open $W \in \mathcal{O}_X$ s.t $\{x_0\}\times Y \subseteq W\times Y \subseteq N$.

Proof. Since $\{x_0\}\times Y \to Y : (x_0, y) \mapsto y$ is Homeo, $\{x_0\}\times Y$ is Compact.
Since N is open, for any $y \in Y$, $\exists\, \mathcal{U}_y \in \mathcal{O}_X$, $V_y \in \mathcal{O}_Y$ s.t $\{x_0\}\times Y \subseteq \mathcal{U}_y \times V_y \subseteq N$.
Now, open cover $\{\mathcal{U}_y \times V_y \mid y \in Y\}$ of $\{x_0\}\times Y$ has a finite subcover

$$\{\mathcal{U}_{y_i}\times V_{y_i} \mid i = 1, \dots, N\}.$$

Set $W = \bigcap_{i=1}^{N} \mathcal{U}_{y_i}$. Then, $\{x_0\}\times Y \subseteq W\times Y$ is clear, and

$$(x,y) \in W\times Y \iff \forall i = 1,\dots,N,\ x \in \mathcal{U}_{y_i} \text{ and } \exists j\ y \in V_{y_j}$$

$$\Rightarrow (x,y) \in \mathcal{U}_{y_j}\times V_{y_j} \subseteq N.$$

「8. Let X be any space, Y be a compact Hausdorff. Then,

$f: X \to Y$ is continuous $\underset{\xleftarrow{\text{compact}}}{\overset{\xrightarrow{T_2}}{\text{if and only if}}}$ $G_f = \{(x, f(x)) \mid x \in X\} \subseteq X \times Y$ is closed.

proof. Suppose that f is continuous. Let $(x,y) \in (X \times Y) \setminus G_f$ be given. Clearly $y \neq f(x)$.
Since Y is T_2, there exist disjoint opens $\mathcal{U}, V \subseteq Y$ containing $y, f(x)$, respectively.
In other words,

$$(y, f(x)) \in \mathcal{U} \times V \subseteq (Y \times Y) \setminus \Delta(Y)$$

And, $f(x) \in V$ implies $x \in f^{-1}[f[\{x\}]] \subseteq f^{-1}[V] \subseteq X$. Now enough to show that only (※).

$$(x,y) \in f^{-1}[V] \times \mathcal{U} \overset{(※)}{\subseteq} (X \times Y) \setminus G_f$$

because f is continuous, thus $f^{-1}[V]$ is open. The inclusion (※) equals to:

$$\left(f^{-1}[V] \times \mathcal{U}\right) \cap G_f = \emptyset$$

and it is true. if it is not empty set, then for some $(x, f(x)) \in G_f$,
$(x, f(x)) \in f^{-1}[V] \times \mathcal{U}$. That is, $f(x) \in V$ and $f(x) \in \mathcal{U}$, which is contradiction.
(Note: in this direction, we used only T_2 condition.)
Conversely, suppose that the graph G_f is closed set.
Let $\mathcal{U} \subseteq Y$ be an open set. Want to show that: $f^{-1}[\mathcal{U}]$ is an open set of X.
Let $x \in f^{-1}[\mathcal{U}]$ be given. Then, $f(x) \in \mathcal{U}$ by definition.
Let $x_0 \in X$ be given. Then, for any open $V \subseteq Y$ containing $f(x_0)$,

$$G_f \cap (X \times (Y \setminus V))$$

is closed set. Now, since Y is compact, the projection $\pi_X : X \times Y \to X$ is closed map, thus

$$\pi_X[G_f \cap (X \times (Y \setminus V))] \subseteq X \text{ is closed in } X.$$

Now, $\mathcal{U} = X \setminus \pi_X[G_f \cap (X \times (Y \setminus V))]$ is open, moreover, $x_0 \in \mathcal{U}$ because

$$x_0 \in \mathcal{U} \iff x_0 \notin \pi_X[G_f \cap (X \times (Y \setminus V))] \iff \forall (x,y) \in G_f \cap (X \times (Y \setminus V)),\ \pi_X(x,y) = x \neq x_0$$

But, since $(x_0, f(x_0)) \notin (X \times (Y \setminus V))$, the last assertion holds. Now, only showing $f[\mathcal{U}] \subseteq V$ is remained.
Let $x \in \mathcal{U}$. Then, $f(x) \notin Y \setminus V$, that is, $f(x) \in V$. ∎

Γ9. The Generalized Tube Lemma.

Let $A \subseteq X$, $B \subseteq Y$ be Compact subspaces. Then,

For any open $N \subseteq X \times Y$ Containing $A \times B$, there exists opens $\mathcal{U} \subseteq X$ and $V \subseteq Y$ s.t $A \times B \subseteq \mathcal{U} \times V \subseteq N$

Proof.

Γ12. Let $p : X \to Y$ be a closed continuous onto map s.t for all $y \in Y$, $p^{-1}[\{y\}]$ is Compact.

If Y is compact, then X is Compact. (Such p is called perfect map.)

Proof. Technically, we can write

$$X = p^{-1}[Y] = p^{-1}\Big[\bigcup_{y \in Y} \{y\}\Big] = \bigcup_{y \in Y} p^{-1}[\{y\}]$$

Let $\mathcal{O} = \{\mathcal{U}_\alpha \mid \alpha \in \Lambda\}$ be an open Cover of X. For each $y \in Y$,

$$p^{-1}[\{y\}] \subseteq \bigcup_{y \in Y} p^{-1}[\{y\}] \subseteq \bigcup_{\alpha \in \Lambda} \mathcal{U}_\alpha$$

And by the assumption, $p^{-1}[\{y\}]$ is Compact, therefore there is a finite index $\Lambda_y \subseteq \Lambda$ s.t

$$p^{-1}[\{y\}] \subseteq \bigcup_{i \in \Lambda_y} \mathcal{U}_i.$$

Now, since p is Closed map,

$$V_y = Y \setminus p\Big[X \setminus \Big(\bigcup_{i \in \Lambda_y} \mathcal{U}_i\Big)\Big]$$

is an open set of Y, Containing y. Now, since Y is compact, for some $y_1, \cdots, y_n$,

$$Y = \bigcup_{y \in Y} V_y = \bigcup_{j=1}^{n} V_{y_j} \Rightarrow X = p^{-1}[Y] = p^{-1}\Big[\bigcup_{j=1}^{n} V_{y_j}\Big] = \bigcup_{j=1}^{n} p^{-1}[V_{y_j}]$$

Meanwhile, $p^{-1}[V_y] = p^{-1}\Big[Y \setminus p\Big[X \setminus \bigcup_{i \in \Lambda_y} \mathcal{U}_i\Big]\Big] = X \setminus p^{-1}\Big[p\Big[X \setminus \bigcup_{i \in \Lambda_y} \mathcal{U}_i\Big]\Big] \subseteq \bigcup_{i \in \Lambda_y} \mathcal{U}_i$,

therefore $Y \subseteq \bigcup_{j=1}^{n} \bigcup_{i \in \Lambda_{y_j}} \mathcal{U}_i$

「+ Extra problem

#1. Let X be a Hausdorff space, and $K_\alpha \subseteq X$ $(\alpha \in \Lambda)$ be Compact subsets. If $\{K_\alpha \mid \alpha \in \Lambda\}$ satisfies F.I.P., then $\bigcap_{\alpha \in \Lambda} K_\alpha \neq \phi$.

Proof. Suppose that $\bigcap_{\alpha\in\Lambda} K_\alpha = \phi$. Let $\beta \in \Lambda$ be fixed. Then,

$$K_\beta \cap \Big(\bigcap_{\alpha \neq \beta} K_\alpha\Big) = \bigcap_{\alpha \in \Lambda} K_\alpha = \phi, \text{ thus } K_\beta \subseteq \bigcup_{\alpha \neq \beta} K_\alpha^c$$

Since X is T_2, K_α^c is open, so $\{K_\alpha^c \mid \alpha \in \Lambda \setminus \{\beta\}\}$ is open cover of K_β. Now, there is a finite subcover s.t

$$K_\beta \subseteq \bigcup_{i=1}^{n} K_{\alpha_i}^c, \text{ that is, } K_\beta \cap \Big(\bigcap_{i=1}^{n} K_{\alpha_i}\Big) = \phi.$$

It is Contradiction to F.I.P.

#2. Let X be a Compact set, and $A \subseteq X$ be an infinite subset. Then, $A' \neq \phi$.

Proof. Suppose that $A' = \phi$. Then, for any $x \in X$, there is an open U_x containing x

$$A \cap U_x \setminus \{x\} = \phi.$$

Since $\{U_x \mid x \in X\}$ is open cover of X, there is a finite subcover $\{U_{x_i} \mid i = 1, 2, \cdots, n\}$.

Then, $A \subseteq X = \bigcup_{i=1}^{n} U_{x_i}$, but $A = A \cap \Big(\bigcup_{i=1}^{n} U_{x_i}\Big) = \bigcup_{i=1}^{n} (A \cap U_{x_i}) \subseteq \bigcup_{i=1}^{n} \{x_i\}$, which is Contradiction.

\# 3. Co-Countable space $(\mathbb{R}, \mathcal{T}_c)$.

In $(\mathbb{R}, \mathcal{T}_c)$, $A \subseteq \mathbb{R}$ is Compact $\Leftrightarrow$ A is finite.

proof. If A is finite, then trivial. If A is infinite, Construse $\{x_n\}$ as $x_1 \in A$, $x_2 \in A \setminus \{x_1\}$, $x_3 \in A \setminus \{x_1, x_2\}$, ---, then,

$\mathcal{U}_n = \mathbb{R} \setminus \{x_i \mid i \in \mathbb{N} \setminus \{n\}\}$ is open of the given space, and

$A \subseteq \bigcup_{n \in \mathbb{N}} \mathcal{U}_n$ but, it has no finite subcover

If $f : \mathbb{R} \to (\mathbb{R}, \mathcal{T}_c)$ is Continuous, it is Constant map.

Proof. Since the Compactness is preserved by Continuous, for any $[a,b]$,

$$f[[a,b]] \text{ is compact in } (\mathbb{R}, \mathcal{T}_c).$$

That is, $f[[a,b]]$ is finite for any $a<b$. Now,

$$f[\mathbb{R}] = f\left[\bigcup_{n \in \mathbb{Z}} [n, n+1]\right] = \bigcup_{n \in \mathbb{Z}} f[[n, n+1]]$$ is at most Countable.

Meanwhile, Since the Closedness is preserved by Continuous Preimage, every finite or Countable subset C, $f^{-1}[C]$ is closed.

Denote $\operatorname{Im} f = \{x_n \mid n \in \mathbb{N}\}$. Let $A = f^{-1}[\{x_1\}]$, $B = f^{-1}[\{x_n \mid n \in \mathbb{N} \setminus \{1\}\}]$.

Then, $\mathbb{R} = A \cup B$ and $A \cap B = \emptyset$. And both A and B are Closed,

$A = \mathbb{R} \setminus B$ is clopen, $A = \mathbb{R}$. Contradiction.

*Fail Try of 12.

⌈12. Let $p: X \to Y$ be a closed continuous onto map s.t for all $y \in Y$, $p^{-1}[\{y\}]$ is Compact.
If Y is compact, then X is Compact. (Such p is called perfect map.)

Proof. Let $\mathcal{O} \subseteq \tau_X$ be an open cover of X. That is, $\bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U} = X$

Now, by the surjectivity, $p\left[\bigcup_{\mathcal{U} \in \mathcal{O}} \mathcal{U}\right] = \bigcup_{\mathcal{U} \in \mathcal{O}} p[\mathcal{U}] = Y$.

And, since p is Closed onto map, it is open map. Therefore, $p[\mathcal{U}]$ $(\mathcal{U} \in \mathcal{O})$ is open cover of Y, thus it has finite subcover s.t $\bigcup_{i=1}^{N} p[\mathcal{U}_i] = Y$, since Y is compact. Observe that

$$X = p^{-1}[Y] = p^{-1}\left[\bigcup_{i=1}^{N} p[\mathcal{U}_i]\right] = \bigcup_{i=1}^{N} p^{-1}[p[\mathcal{U}_i]]$$

Let $y \in Y$ be given. By the surjectivity, $p^{-1}[\{y\}]$ is non-empty.
Let $x_0 \in X$ s.t $p(x_0) = y$.

$$p^{-1}[W] \subseteq \mathcal{U}$$
$$\Longleftrightarrow W \subseteq p[\mathcal{U}]$$

Let $y \in Y$ be given. If $\mathcal{U} \subseteq X$ is open containing $p^{-1}[\{y\}]$, then

$$p^{-1}[\{y\}] \subseteq \mathcal{U} \Rightarrow p[p^{-1}[\{y\}]] \subseteq p[\mathcal{U}] \Rightarrow y \in p[\mathcal{U}]$$
$$\Rightarrow y \in W \subseteq p[\mathcal{U}]$$

$$Y = \bigcup_{y \in Y} \left(Y \setminus p\left[X \setminus \left(\bigcup_{i \in \Lambda_y} \mathcal{U}_i\right)\right]\right) = \bigcup_{j=1}^{n} \left(Y \setminus p\left[X \setminus \left(\bigcup_{i \in \Lambda_{y_j}} \mathcal{U}_i\right)\right]\right)$$